A

Mini Project Report

On

"PREDICTION OF ENERGY SURPLUS & SHORTAGE USING ML"

Submitted for partial fulfillment of requirement for the award of degree

of

Master of Business Administration
(Artificial Intelligence and Data Science)

GRAPHIC ERA (DEEMED TO BE UNIVERSITY)
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DECLARATION

I hereby declare that the Mini Project entitled *"PREDICTION OF ENERGY SURPLUS & SHORTAGE USING ML"* submitted for the Degree of Master of Business Administration in Artificial Intelligence and Data Science, is my original work and the Mini Project has not formed the basis for the award of any degree, diploma, associateship, fellowship or similar other titles. It has not been submitted to any other University or Institution for the award of any degree or diploma.

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ACKNOWLEDGEMENT

I express my sincere thanks to my project guide, Mr./Dr./Ms./Mrs. ______________________ Designation __________________, Department ______________________ for guiding me right form the inception till the successful completion of the project.

I also record my indebtedness to my supervisor, Prof. /Dr. /Mr. /Ms. ____________________________, for his / her counsel and guidance during the preparation of this Mini Project. I am grateful to (Director, Head or any other faculty) ……………………………………………………

I wish to record my sincere thanks to (your family members or friends etc.) .... for their help and cooperation throughout our project. My thanks are due to (those who have helped in collecting data or analysis or typesetting etc.).,

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ABSTRACT

This project addresses the critical challenge of predicting energy shortages and surpluses in the Indian state of Uttarakhand using machine learning (ML) techniques. Given the state's heavy reliance on hydroelectric power, coupled with its geographical and seasonal variability, energy supply often fails to align with dynamic and growing demands. This leads to frequent power shortages that disrupt socio-economic activities and, conversely, to surplus generation that results in resource wastage. The study aims to develop a predictive modelling framework using historical energy data to enable data-driven decision-making in energy procurement and load management.

The approach involves leveraging daily historical records including energy demand, supply, drawl from Indian Energy Exchange (IEX) and Power Exchange India Ltd (PXIL), banking transactions, and calculated shortage/surplus values. The dataset spans approximately three years and is processed to extract temporal patterns, rolling statistics, and lag-based features. These features form the input for a multi-model ML pipeline involving regression, classification, and time-series forecasting techniques.

Three major types of ML models were employed. Regression models—especially XGBoost Regression—were effective in predicting the exact magnitude of shortages or surpluses, achieving high accuracy with an R^2 score of approximately 0.89 and a Root Mean Squared Error (RMSE) of ~2.3 MW. Classification models, particularly Random Forest Classifier, provided accurate alerts by distinguishing between shortage and surplus days, with an F1-score close to 0.90, thereby supporting timely operational decisions. Time-series models like Long Short-Term Memory (LSTM) networks proved particularly effective in capturing long-term dependencies in demand and supply fluctuations, outperforming ARIMA in long-range forecasting with a Mean Absolute Percentage Error (MAPE) of around 4–6%.

The results were validated using multiple techniques including k-fold cross-validation, train-test split analysis, walk-forward validation, and hyperparameter tuning. The models demonstrated strong alignment with real-world seasonal patterns and demand cycles observed in Uttarakhand's energy ecosystem. Insights gained during exploratory data analysis (EDA) and model interpretation highlighted the impact of temporal features (e.g., month, day of the week) and market-based energy drawl mechanisms on shortage prediction.

The outcomes of this project have significant practical implications. The models can be integrated into energy planning systems to generate daily alerts, support procurement planning, and aid long-term policy decisions. However, the project also faced limitations such as the exclusion of real-time weather, political, or infrastructural variables, and the reliance on historical static datasets without live streaming capabilities. Computational complexity, especially in LSTM training, was another challenge in terms of real-time deployment readiness.

Looking ahead, the study lays a strong foundation for future enhancements such as multi-step forecasting, real-time model updates with streaming data, and integration with cloud-based APIs and dashboards. These improvements could evolve the current framework into a fully-fledged intelligent energy management system capable of supporting smart grid infrastructure and contributing to India's energy sustainability goals.

CHATPER:1 INTRODUCTION

Over the last few years, increasing complexity in power distribution grids and varying power needs have created a serious problem for regional power distribution corporations and electricity boards. In a geographically complex and diverse state such as Uttarakhand, ensuring the balance between supply and demand of energy has become more challenging. Power shortages become a routine hassle in day-to-day life, impact economic productivity, and burden the existing infrastructure, while power surpluses lead to wastage and economic losses. The issues urgently demand smart and proactive energy management systems. This project meets this challenge by using Machine Learning (ML) algorithms to forecast energy shortages and surpluses based on past data and other relevant external variables.

The prime concern of the present study is the application of historical energy demand, generation, and market transaction data of the state of Uttarakhand in developing forecasting models that can effectively forecast energy shortage. The forecasting capability will be utilized to enable energy management institutions to make evidence-based decisions, reduce energy purchase, and implement pre-emptive actions in coping with shortage.

Machine Learning, being a part of Artificial Intelligence, offers an impressive set of tools that have the capability of finding patterns, fitting non-linear models, and making accurate predictions from large, complex data. Supervised models of learning like regression and classification models and time-series forecasting models are utilized within the project in order to accomplish the task of prediction. The study aims at building a powerful and flexible system of forecasting with model training of features such as daily energy demand, supply measures, banking power trades, and exchange drawl data.

The predictive modeling approach also serves higher goals of enhanced energy efficiency and sustainability. As increasing energy demands and climatic volatility continue to impact hydro-based energy sources in Uttarakhand, predictive information becomes more vital to maintaining grid stability. Furthermore, with increasing emphasis in India on smart grid technology and digital energy management, integrating ML-based forecasting into operations is a timely development.

This introduction provides the context for a detailed discussion of the following issues: domain knowledge of energy system in Uttarakhand, the use and promise of AI and ML in contemporary analytics, the real application of ML for energy forecasting, and an extensive survey of appropriate ML algorithms. Combined, these papers will illustrate how data-driven methodologies can play a critical role in addressing real-world energy management problems.

1.1. DOMAIN SPECIFIC KNOWLEDGE

The energy sector is the engine of every economy and facilitates growth, productivity, and social advancement. In this sector, there needs to be a steady and stable supply of electricity. Energy management keeps generation, transmission, distribution, and consumption in balance to meet varying demand, minimizing loss and wastage. The Indian state of Uttarakhand offers unique challenges in this regard due to its geographical, infrastructural, and environmental characteristics.

Uttarakhand, being a mountainous state with a dispersed population, depends predominantly on hydroelectricity. Though the green source of energy is eco-friendly, it introduces uncertainty to the value chain of energy. Hydroelectric power generation is a seasonal process, and monsoons and winter dominate the process. Lean months are the months when water levels in dams decrease and hence less electricity is generated. Monsoons are the months when heavy rainfall can generate excess power, which could or could not be equal to demand.

1.2. Artificial Intelligence & Machine Learning Technologies

Artificial Intelligence (AI) and its subset, Machine Learning (ML), have transformed problem-solving into every industry nearly everywhere. Essentially, AI is the development of systems that can simulate human intelligence, learn from experience, and make choices. Machine Learning is an algorithm that teaches us to perform a task better with experience from data over time—without being programmed.

ML models operate by learning the intrinsic patterns and relationships between input variables (features) and output (target) from historical data. These algorithms can handle high-dimensional data, handle noise or missing data, and learn from novel data. Unlike rule-based systems, ML models don't employ pre-defined rules but learn generalizations from examples and are therefore well-suited for dynamic environments like the energy industry. Advancements in ML today are complemented by the development of deep learning, which is based on the use of artificial neural networks to identify complex patterns in large data sets. Deep learning techniques like Long Short-Term Memory (LSTM) networks are best suited to handle time-series data—data ordered in time indexing, like energy demand hourly or generation data by the day.

In addition to prediction, ML also facilitates anomaly detection, classification, clustering, optimization, and recommendation, which can be applied to energy forecasting, demand-side management, asset maintenance, and fraud detection.

The technology stack used in ML includes programming languages like Python and tools like Jupiter Notebooks for experimentation. Libraries like Pandas and NumPy assist with data manipulation, while Scikit-learn, XGBoost, and TensorFlow provide the building blocks for model building and deployment. Visualization tools like Matplotlib and Power BI assist with the easy interpretation of results and insights.

1.3. Applying Machine Learning to the Domain

Using ML in the energy industry, particularly to forecast shortages and surpluses, is to develop the supply-demand system based on historical and current data. The concept is to develop a forecasting system that will alert the authorities of a potential imbalance between demand and supply of energy days or weeks ahead.

Data preprocessing in the form of cleaning and feature transformation of raw data is the initial step. Energy data in this project for Uttarakhand is raw data of day-to-day energy drawl, availability, drawl from exchanges (IEX/PXIL), and banking transactions. Contextual data like time (month, day of week) and weather conditions, if available, are also utilized. Then exploratory data analysis (EDA) is employed to seek patterns and relationships. Visualization indicates that shortages occur more frequently than surpluses and demand typically outstrip availability. By analyzing the lag effect in patterns of demand, time-based patterns such as weekends, holidays, and seasons can be incorporated.

Then, feature engineering is employed to build new variables that better represent the underlying processes. These include lag-based features (e.g., yesterday's demand), rolling averages, and categorical encodings of time-based features.

These ML models are trained using this rich feature dataset. The model is trained in past and current variables as inputs and produces shortage or surplus of energy as outputs. Depending on the formulation, this could either be a regression problem (forecasting the level of shortage/surplus in megawatts) or a classification problem (forecasting labels like "Shortage," "Balanced," or "Surplus").

Regression models help to estimate the forecasted shortage, a critical factor in planning imports of energy or grid balancing of loads. Classification models help in categorizing risk levels in facilitating quick operations. ARIMA and LSTM type time-series models also help with prediction by building an insight into patterns over time.

The true strength of ML here is that it can learn and adapt based on new data. As market dynamics evolve or the patterns of demand change, retraining the model with fresh data ensures it remains accurate and relevant. These models can be incorporated into the dashboards of energy managers to give not just predictions but also actionable recommendations on procurement and load allocation.

1.4. Machine Learning Algorithms

Selecting the right ML algorithms is of paramount significance to the performance of the model. Shortfalls and surpluses of energy can be forecasted in three manners—regression, classification, and time-series forecasting—depending on the nature of the problem and data.

1.4.1. Regression Models

These are appropriate when the question is to estimate the numerical amount of the excess or deficit of energy:

- Linear Regression is the simplest model. It makes the linear relationship assumption between target and input variables (e.g., banking power, energy demand). Simple and easy to comprehend, though, it won't do a good job on complex, nonlinear data.
- There is Random Forest Regression, an ensemble method that generates numerous decisions trees and averages their output. It can model nonlinear interactions between features and is robust to noise.
- XGBoost Regression (Extreme Gradient Boosting) is an efficient algorithm that applies gradient boosting to build progressively better predictions. It performs particularly well with structured tabular data like energy records.

1.4.2. Classification Models

It is handy if we set the energy status (e.g., Shortage = 1, Surplus = 0):

- Logistic Regression provides probabilistic outputs for binary classification tasks. It is simple and works well when the classes are linearly separable.

- Random Forest Classifier is a classification extension of Random Forest. It is suitable for high-dimensional data and provides feature importance values.
- Support Vector Machines (SVM) are useful in determining the best decision boundaries, particularly when there are well-defined margins among classes.

1.4.3. Time-Series Forecasting Models

These models capture temporal dependencies and trends in the data:

- ARIMA (Autoregressive Integrated Moving Average) fits a univariate time series and is appropriate when data is stationary or rendered stationary.
- LSTM (Long Short-Term Memory) networks are RNN variations with which it is now possible to learn long-term dependencies in sequence data quite well. They are particularly well suited to model day or hour energy demand fluctuations.

All these algorithms have advantages and disadvantages. Usually, several models are trained and compared against measures such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), R^2 score (in the context of regression) or accuracy, precision, recall, and F1-score (in the context of classification).

Using this set of models, the project aims to have an accurate, scalable, and deployable predictive model that can be retrained and refitted with new features on a regular basis as the data evolves.

1.5. Model Used

In this research the predictive modeling strategy utilized the Random Forest algorithm using Random Forest Regressor or Random Forest Classified depending on whether the prediction was based on continuous or categorical outcomes. Random Forest is a type of ensemble learning that builds a multitude of decision trees and integrates their predictions to improve your prediction accuracy and minimize overfitting. Given the complexities and nonlinear patterns often present in energy consumption and production, this is a good method for these types of modelling approaches. For the continuous or regression modelling tasks (attempting to predict future energy demand / demand forecast or surplus / energy surplus), the model utilized the Random Forest Regression algorithm. Using historical numeric data, the model makes an output future value prediction from the input of multiple independent ~ features. For categorical modelling tasks (e.g., energy shortage / surplus classification), the behaviors and patterns of the input data for the feature finally output through Random Forest Classifier which provided the binary or categorical results.

The Random Forest was chosen as it is robust enough to deal with the problem at hand, manage both numerical and categorical data, and perform well with noisy/missing data. In addition, Random Forest sheds light on the importance of features, which can lead to identification of important drivers in energy fluctuations predicating shortage/surpluses. Data Preprocessing consisted of normalizing and engineering data into features in order to make the model effective and efficient. Overall, the Random Forest was chosen as it identified a good predictive power to interpretability balance and that it was a good candidate for modeling shortages and surpluses of energy. The Random Forest can also address a variety of types of problems, adding value to this analysis, as it can engage in short-term forecasting, as well as strategic planning for demand management. Model performance is based on R^2 score, mean squared error for regression and accuracy and classification report on classification problems.

CHAPTER:2 LITRATURE REVIEW

2.1. Conventional Storage Technologies

2.1.1. Pumped Hydro and Compressed Air Energy Storage

2.1.1.1. Pumped Hydro Energy Storage:

The study on pumped hydro energy storage presents a very efficient and optimal dispatch model. The model successfully integrates renewable energy sources, namely wind energy and solar photovoltaic power, with the functionality of hydro storage systems. Using mixed-integer nonlinear programming (MINLP), in conjunction with an in-depth case study that considers different electricity market scenarios in Spain in the years 2023 and 2030, the study easily depicts the mechanisms through which energy can be strategically exported and imported. This strategic mechanism is founded on varying price signals within the market, whose ultimate intent is to maximize the generation of green hydrogen using the electrolysis process. *(Tipán-Salazar, Naval, & Yusta, 2025)*

2.1.1.2. Compressed Air Energy Storage (CAES):

The other research deals with CAES in geological formations as well as with renewable generation. It analyses the effect of wind integration and wind forecasting errors by applying various optimization techniques (SQP, ABC, AGTO) to reduce cost imbalances and improve voltage stability. The two research papers do identify, however, that although economic feasibility is considered, issues such as environmental factors, uncertainty modelling, and large-scale operational testing are underdeveloped.
(M. P. Kumar et al., 2025)

2.2. Emerging and Alternative Storage Technologies

2.2.1. Hydrogen-Based Storage Solutions

2.2.1.1. Hydrogen Storage in Micro-Grids:

Specific research focuses on the complex design of a micro-grid specifically for near Zero Energy Buildings, or nZEBs. Hydrogen within this new micro-grid is generated through the process of electrolysis, utilizing excess solar power that is created. The overall purpose of this design is to store energy over the long term, thus making the system efficient enough to survive in grid isolation. This is accomplished through a successfully concerted interface with external hydrogen networks for plug-and-play capability while still being self-sufficient. *(Stamatellos & Stamatellou, 2024)*

2.2.1.2. Underground Hydrogen Storage (UHS):

Added to the above, yet another survey assesses underground hydrogen storage for India's net-zero targets. It assesses geological amenability, hydrogen production, storage choices, and the general policy framework, albeit it requires further detailed economic and monitoring study. *(S. Kumar et al., 2023)*

2.2.2. Thermal and Phase Change Energy Storage

2.2.2.1. Thermal Storage with Machine Learning:

In the context of studying cogeneration plants and more so those that are classified under the combined heat and power systems (CHP) category, a certain study utilizes advanced machine learning techniques to enhance the optimization of thermal storage application. By using methods to accurately forecast short-term thermal loads, the study shows a considerable drop in the amount of storage capacity needed. It is, however, worth noting that the findings of this study are specific to a certain greenhouse environment, which may limit the findings' generalizability. *(Ghilardi et al., 2023)*

2.2.2.2. Carnot Batteries with Enhanced Phase Change Materials:

A similar study suggests a new and advanced Compressed Heat Energy Storage, or CHEST system, as it is also referred to. This system employs cascading phase change materials, or PCMs, that significantly enhance the round-trip efficiency and energy density of the storage system. However, it must be noted that this study is still largely theoretical in nature at this juncture, and it emphasizes the absolute necessity of further tests with regard to cost-effectiveness and durability in actual applications. *(Tafone, Pili, Pihl Andersen, & Romagnoli, 2023)*

2.2.3. Other Alternative Technologies

2.2.3.1. Liquid Air Energy Storage (LAES):

While LAES demonstrated a round-trip efficiency above 60% using a Solvay cycle, non-linear heat capacity profiles, and finding pressure conditions, the work fails to compare any other storage technologies. *(Chaitanya, Narasimhan, & Venkatarathnam, 2023)*

2.2.3.2. Seawater Battery (SWB):

Following an investigation of the potential of seawater batteries (SWBs), they can deal with intermittent renewable energy. Furthermore, multiple charging scenarios suggest the seawater battery produces high energy efficiency. The study, in addition to also exploring various charging scenarios and using a predictive model (LSTM) to improve performance, the study acknowledges uncertainties that could impact implementation due to issues relating to corrosion or scale-up. *(Park et al., 2023)*

2.3. Battery Energy Storage Systems (BESS) and Related Approaches

2.3.1. Conventional BESS Optimization

2.3.1.1. BESS Sizing and Degradation:

The review article on BESS provides thorough discussions focusing on issues related to the intermittent renewable generation and balancing supply-demand requirements while dealing with battery degradation. The article also provides recommendations for batteries design and capacity cost planning. The article is mainly a synthesis of literature and does not really address new simulation approaches. *(Talebi & Aly, 2024)*

2.3.1.2. BESS Deployment with Solar Integration:

As a complementary work, the authors provide an analysis of optimal location and sizing of BESS (battery energy storage systems) owned by customers or utilities, specifically for operation on a solar power farm. They performed analysis using multi-objective and soft-margin classification and show the effect that the interaction of network topology has on the system's net efficiency and energy captured in surplus energy from the sun. The work does not provide solutions to the inherent problems of governmental regulation, or dynamic physical operational issues that arise from regulation. *(Mannepalli et al., 2022)*

2.3.2. Second-life Battery Application

2.3.2.1. Reusing Car Batteries for Wind Farms:

This paper investigates the possibility of using second-life batteries from electric vehicles in a wind farm biomass setting on the island of the study, addressing energy system waste and circular economy principles. The author also highlights that their modelling and techno-economic assessment of the second-life battery showed financial and environmental feasibility, but more research is needed to account for degradation and safety specifications. *(López, Ramírez-Díaz, Castilla-Rodríguez, Gurriarán, & Mendez-Perez, 2023)*

2.4. Integrated Energy System and Grid Applications

2.4.1. Grid Stability and Integrated Planning

2.4.1.1. DG and ESS Planning:

Research conducted on Distributed Generation (DG) and Energy Storage Systems (ESS) planning proposes optimal siting while employing vulnerability assessment and multi-objective optimization on a typical IEEE test system. This highlighted advantages of better profitability and lower costs, however, there is little real-world validation of the above work. *(Yang, Teh, & Alharbi, 2024)*

2.4.1.2. Energy Storage for Grid Stability:

Another research study focuses on balancing energy surpluses and deficits to encourage long-term stability of the electrical grid. Although the study does model cycles of charging/discharging for seasonal storage and looks to balance energy surpluses and deficits, the research is not furthered in testing border fault scenarios or understanding their limit to electrical grid stability. *(Gwozdz & Przygrodzki, 2024)*

2.4.1.3. Optimizing the Energy Storage Mix:

In this analysis, the ideal combination of storage technologies is derived based on predicted annual surplus renewable energy. While the analysis incorporates system constraints and annual surplus pathway, it does highlight challenges around technology readiness for commercialization (e.g., relative TRLs) and policy. *(Lee, Jung, Lee, & Jang, 2023)*

2.4.2. Evaluation Metrics and Market Dynamics

2.4.2.1. ESS Efficiency Metrics:

After reviewing the metrics associated with system efficiency in high-renewable microgrids, the following new evaluation metrics are proposed: Loss of Surplus Energy Rate (LSER), Expected Loss of Surplus Energy (ELSE). These metrics are intended to provide approaches to adapt system sizing to provide efficiency beyond the consideration of "cost" alternatives. *(Karimi Ghaleh Jough & Gholami, 2024)*

2.4.2.2. Energy Trading and Market Power:

There are two studies that cover the market side of storage:

- One provides an energy trading mechanism for industrial parks with carbon constraints. *(Zhu et al., 2024)*
- The other considers how storage and generation market power impacts wholesale market outcomes in Britain and suggests more localized analyses. *(Williams & Green, 2022)*

2.4.2.3. Economic Analysis of Power Shortages:

A more extensive study of the economy employs a Computable General Equilibrium (CGE) model to measure the total effect of power shortages, across many sectors, in China. While it provides meaningful sector-specific insights, also noted in the prefaced report from the State Electricity Regulatory Commission of China, it highlighted the need to develop improved real-time models and models responsive to policy changes. *(Ou, Huang, & Yao, 2016)*

2.5. Distributed and Consumer-Level Applications

2.5.1. Prosumers and Hybrid System

2.5.1.1. Prosumers in Renewable Energy:

This review identifies the dual roles of prosumers (producers and consumers of renewable energy) and hybrid storage models to support small\-scale residential and community level energy systems. possible socio-economic implications. There is a discussion of areas needing further research into business models and technical combinations. *(Review excerpt; details partially truncated in source)*

2.5.2. Electric Vehicle (EV) Infrastructure

2.5.2.1. EV Charger Deployment:

This project develops a multi-objective planning model to determine the optimal number of EV chargers at renewable energy facilities and the most appropriate placement of the chargers. The model demonstrates how cost, sustainability, and customer demand can be considered together using a multi-criteria Pareto analysis approach. It also demonstrates a first step towards developing a more comprehensive multi-objective planning model for electric vehicle infrastructure while acknowledging the simplification of dynamic pricing & charging behaviours. *(Suzuki & Tanaka, 2024)*

2.5.2.2. Fast Charging Stations Integration:

Related to this, previous studies have attempted to include energy storage systems (e.g., Lithium-Ion Batteries, Vanadium Redox Flow, and hybrid systems with electrolyzes) into fast charging stations. The outcomes from the simulation indicate a possibility of a significant decrease in load on the electrical grid, greater power density, and other usable results, but there is still a knowledge gap when looking at material degradation and total lifecycle costs. *(Pelosi, Longo, Bidini, Zaninelli, & Barelli, 2023)*

CHAPTER:3 DATA AND METHODOLOGY

3.1. Overview

This project involves developing intelligent models for predicting shortages and surpluses of energy in Uttarakhand, using historical energy-related data. The primary goal is to support power distribution companies in making better decisions to minimize procurement costs and maximize the availability of energy. The problem's diverse nature—ranging from continuous prediction to binary classification and time-dependent predictions—calls for a blend of machine learning models.

The methodology is data-driven, commencing from preprocessing to clean the dataset and get it prepared for analysis. EDA revealed solid correlations between demand, availability, and imports from the market. Feature engineering caught temporal dynamics, while model selection focused on interpretability, accuracy, and generalizability.

By benchmarking against traditional models like Linear Regression and ensemble models (Random Forest, XGBoost) and deep learning models (LSTM), the study lands on the best compromise between interpretability and precision. The resulting product is a prediction system that can classify or forecast energy states, thereby supporting smart energy purchasing strategies

3.2. The Dataset's Examples

The dataset, "Energy Dataset.csv," includes daily data with features such as:

• Date
• Energy Availability (MW)
• Energy Demand (MW)
• Withdrawal from PXIL and IEX (MW)
• Banking Power Received (MW)
• Surplus/Shortage Indicator (MW)

Table 1 Examples of Data Entries:

Date	Availability	Demand	PXIL Drawl	IEX Drawl	Banking Power	Shortage/Surplus
2022-01-01	4200	4500	150	100	80	-120
2022-01-02	4300	4400	160	90	70	120

Each row is a daily energy profile, and it is on this that the pattern recognition in the ML models depends. The features are seasonal, temporal, and structural relations that are of most value in the forecasting.

3.3. Data characteristics & Format

The data is in CSV format and structured with unique column headers as well as time-series indexed by date. Features are numeric (e.g., energy in MW) or datetime. There is no missing value or duplication as determined during preprocessing.

Key features are:

3.3.1. Temporal Nature

The Date column anchors each record, making it suitable for time-series modelling.

3.3.2. Numerical Columns

Simple predictors such as Demand, Availability, PXIL, IEX, and Banking Power are continuous.

3.3.3. Computed Feature

Shortage/Surplus column is calculated as Availability - Demand + Imports, which depicts day-to-day energy status.

The structure offers ML pipeline support and supports supervised learning and sequential forecasting tasks.

3.4. Total Dataset Size

The dataset includes about 1095 records (3 years' worth of daily data), which is almost equivalent to 3 complete years of historical energy data for Uttarakhand. Each record refers to one day and it contains approximately 7 remaining columns of useful features after preprocessing and feature engineering. In terms of file size, it is small (approximately 100 KB), but it is packed full of information. The amount of data is sufficient for traditional ML models and small-scale deep learning experiments such as LSTM with the right sequence framing.

3.4.1. Sample Data Entries

Table 2 Sample table representing five entries from the dataset:

Date	Availability (MW)	Demand (MW)	PXIL Drawl (MW)	IEX Drawl (MW)	Banking Power (MW)	Surplus/Shortage (MW)
2022-01-01	4200	4500	150	100	80	-120
2022-01-02	4300	4400	160	90	70	120
2022-01-03	4100	4600	130	110	60	-200
2022-01-04	4400	4300	140	95	85	190
2022-01-05	4250	4450	155	105	75	35

3.5. Model Discussion

3.5.1. About The Algorithms

Given the difficult task of forecasting energy shortages and surpluses in Uttarakhand, a hybrid group of regression, classification, and time-series forecasting models have been used. Each model type is used for a specific prediction type. Regression models (e.g. Linear

Regression, Random Forest Regression, XGBoost Regression) are used to predict the magnitude of shortage/surplus (continuous value). Classification models (e.g. Logistic Regression, Random Forest Classifier, Support Vector Machine (SVM)), are used to classify days into "shortage" or "surplus". Finally, when incorporating temporal dependencies and forecasting future values, time-series forecasting models (e.g. ARIMA and LSTM), will also be applied. These various models allow comparison of the accuracy and actionable intelligence they provide for energy management.

3.5.2. How Will It Work

The first stage of the prediction pipeline consists of data preprocessing of the energy dataset: cleaning, standardizing, and normalizing. The next stage is EDA (exploratory data analysis) to look for possible patterns, trends, and correlations, specifically between energy availability, demand, imports, and actual surpluses or shortages of energy. Following the EDA, feature engineering is applied, to add context variables like month, day of week, and lagged-based variables that might help the model make better predictions. The final stage of data preprocessing includes splitting the data into testing and training sets. Then depending on the framing of the problem a regression, classification, or time-series based model is trained and evaluated based on a set of metrics - for example RMSE, R^2, accuracy, and F1 score. Hyperparameters are then tuned to improve performance of the model. LSTMs need the sequence data to be reshaped accordingly and are trained using a GPU-enabled environment for efficiency.

3.5.3. Mathematics

3.5.3.1. Linear Regression

It develops the line (or hyper-plane) that minimizes the Residual sum of squares between the expected value and the predicted value.

Formula ➔ $\hat{\boldsymbol{y}} = \boldsymbol{\beta_0} + \boldsymbol{\beta_1 x_1} + \boldsymbol{\beta_2 x_2} + \ldots\ldots + \boldsymbol{\beta_n x_n}$

3.5.3.2. Random Forest

It structures numerous decision trees and each tree votes. For regression it takes the mean of all the outputs, for classification it takes a majority vote.

Formula➔ $\hat{\boldsymbol{y}} = \frac{\boldsymbol{1}}{\boldsymbol{T}}\sum_{\boldsymbol{t=1}}^{\boldsymbol{T}} \boldsymbol{h_t}\,(\boldsymbol{x})$

3.5.3.3. XGBoost

It structures a multi-tree object/input, one tree at a time while it descends downwards with the gradient method and addressing a regularized, graded objective function/learning algorithm to achieve the optimal solution.

Formula➔ Minimizes a regularized objective:

$$\boldsymbol{L} = \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{l}(\boldsymbol{y_i}, \widehat{\boldsymbol{y_i}^{(\boldsymbol{t})}}) + \sum_{\boldsymbol{k=1}}^{\boldsymbol{t}} \boldsymbol{\Omega}(\boldsymbol{f_x})$$

with:

$$\boldsymbol{\Omega(f) = \gamma T + \frac{1}{2}\pi \sum_{j=1}^{T} w_j^2}$$

3.5.3.4. Logistics Regression

It models the probability with a sigmoid structure.

Formula ➔ $\boldsymbol{P(y = 1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\ldots\ldots+\beta_n x_n)}}}$

3.5.3.5. SVM

It maximizes the margin found in the hyper-plane between the 2 classes.

Formula ➔ Decision boundary is defined as:

$$\boldsymbol{f(x) = w^T x + b}$$

And the optimization goal:

$\boldsymbol{min \frac{1}{2} \, || \, w|| }$ **subject to** $\boldsymbol{y_i(w^T x_i + b) \geq 1}$

3.5.3.6. ARIMA

It applies lagged values of the past and applied lagged models of the errors.

Formula ➔ $\boldsymbol{y_{t=c+\emptyset_1 y_t - 1 \ldots\ldots + \emptyset_p y_t - p + \theta_{1\epsilon_t} - 1 + \cdots \ldots\ldots + \theta_{q\epsilon_{t-q}} + \epsilon_t}}$

3.5.3.7. LSTM

Its networks can learn long-term dependencies in a sequence because it uses an input gate, a forget gate, and an output gate.

Formula ➔

Forget gate➔ $\boldsymbol{f_{t=\sigma(W_f * [h_{t-1,x_1}]+b_f)}}$

Input gate➔ $\boldsymbol{i_{j=\sigma(\,W_t\,*}\left[h_{t-1,x_t}\right] + b_i)}$

$$\boldsymbol{\overline{C} = \tanh\left(W_c * \left[h_{t-1,x_t}\right] + b_c\right)}$$

Cell state update➔ $\boldsymbol{C_{t=f_{t*}} C_{t-1+} \; i_{t*}\overline{C}_t}$

Output gate➔ $\boldsymbol{o_{t=\sigma}\left(W_o * \left[h_{t-1,x_t}\right] + b_o\right)}$

$$\boldsymbol{h_t = o_{t*} tanh(C_t)}$$

3.5.4. Architecture

The architecture changes based on the model:

3.5.4.1. Regression & Classification Models:

Are built as pipelined data preprocessing → feature selection → train model → validate

3.5.4.2. Random Forest/XGBoost:

Ensemble architectures derived by different trees or groups of multiple working trees.

3.5.4.3. LSTM Architecture:

A deep neural network directed by multiple stacked LSTM cells - each shaped as (samples, timesteps, features), and the output goes through a dense layer.

3.5.4.4. ARIMA:

A statistical model based on lag and differencing order set by a combination of ACF/PACF and AIC. Each model is built with Python based on the complexity and what libraries are required, including Scikit-learn, XGBoost, and TensorFlow.

CHAPTER:4 DEVELOPMENT OF THE MODULE

4.1. Dimensionality Of the Model

The analysed dataset included records of daily measured energy demand vs. energy availability, energy surplus/shortage, energy purchases in IEX/PXIL markets, and banking transactions from Uttarakhand. After cleaning and parity, we modified the dataset to include engineered features to add to the dimensions of the dataset while maintaining relevance. Original features contained features like Demand, Availability, Shortage/Surplus, etc. From Feature Engineering, we finally came to some temporal features (such as month, day of week, is weekend), some lag-based (such as previous day demand, previous week availability), and some differential features (such as demand availability diff) which took the dimensionality to about 12–15 optimized features.

Redundant or strongly correlated features were trimmed using tree-based models' feature importance ranking as well as correlation analysis. All numerical features were scaled, which effectively helped model convergence and stability as well as a better fit for algorithms like logistic regression and long short-term memory (LSTM). The final dataset had features with the necessary temporal, transactional, and contextual richness for robust predictive modelling.

4.2. Framework Of The model

The modelling framework adopted a modular pipeline:

4.2.1. Data Preprocessing:

This involved cleaning up and structuring the dataset. Specifically, date values were converted to datetime objects, and all the critical features were normalized or encoded.

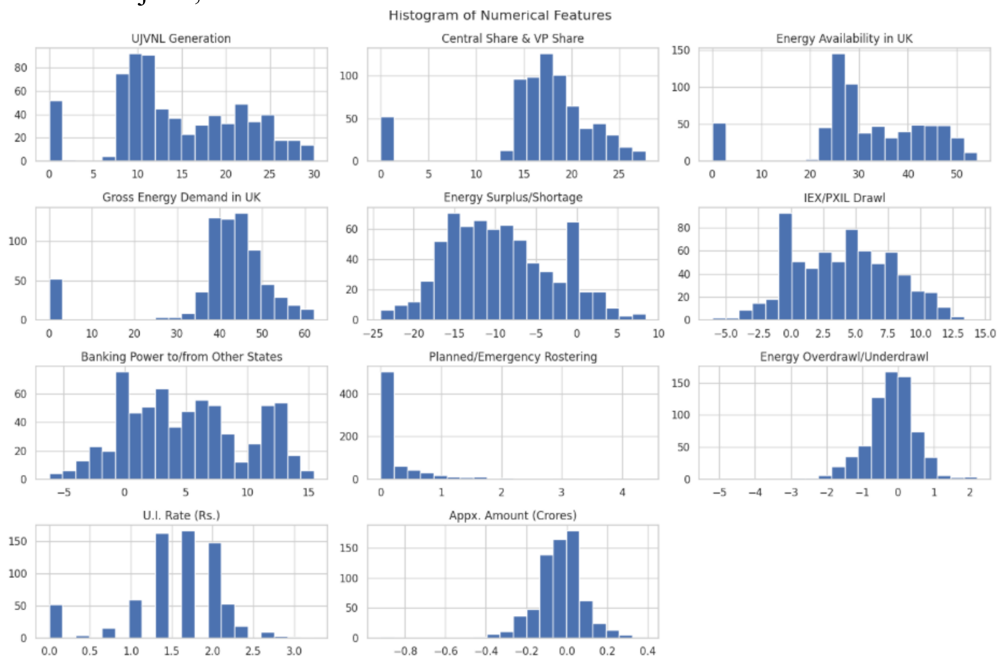

Figure 1 :Histogram showing the distribution of daily energy consumption values.

4.2.2. Feature Engineering:

Generated temporal (e.g., month, day) and lagged features to summarize any recurrent patterns with temporal characteristics around supply-demand variability.

4.2.3. EDA:

Revealed and visualized persistent shortages, peak demand periods, and strong correlations between demand and availability.

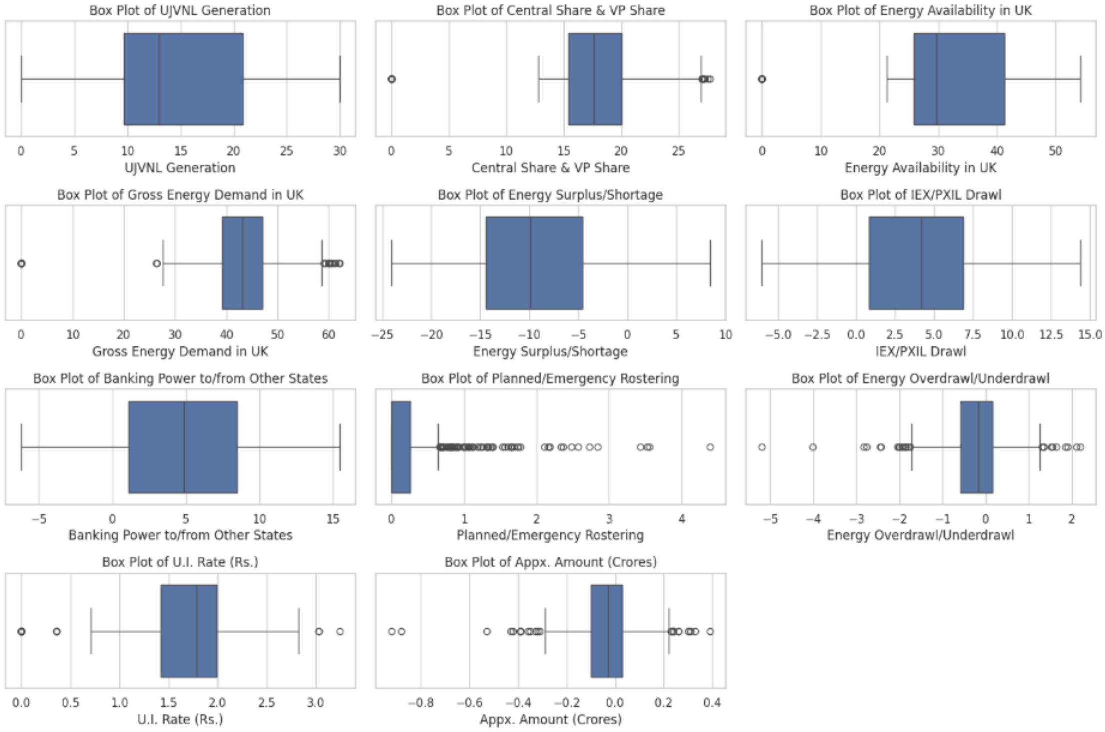

Figure 2 : Box plot highlighting outliers and distribution across key energy features.

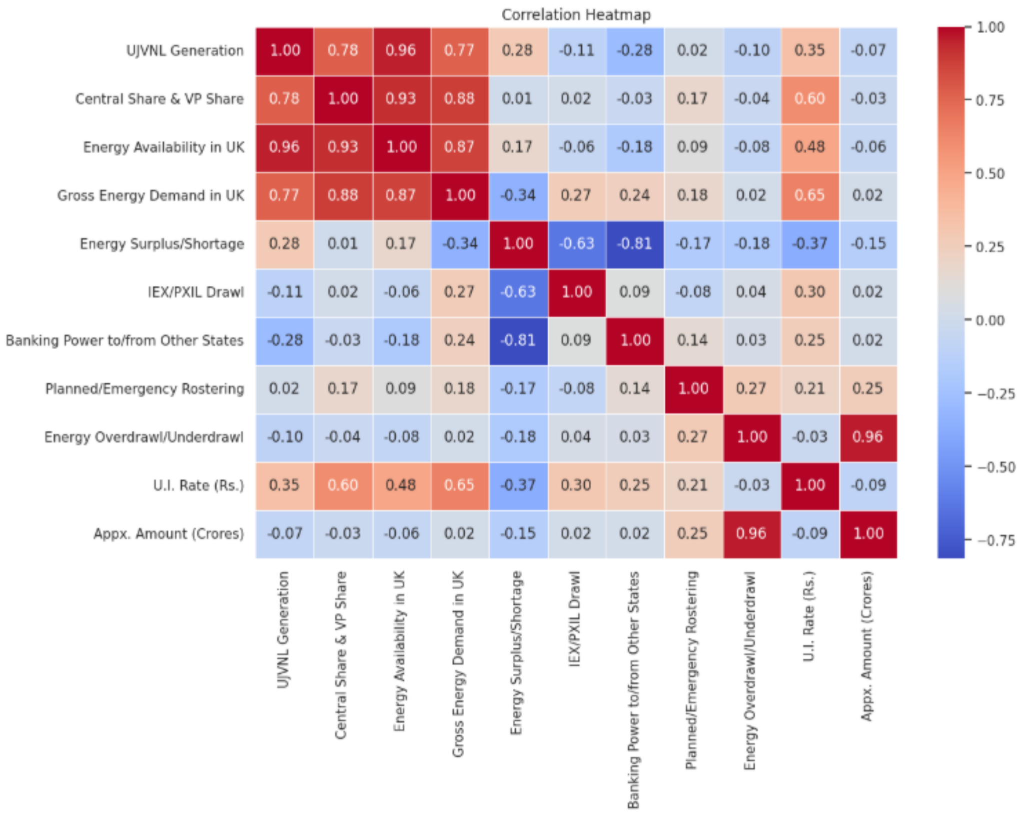

Figure 3 : Correlation Heatmap of Energy Variables in Uttarakhand.

Figure 4 : Pair plot showing pairwise relationships among selected numerical features.

4.2.4. Model Training:

Models were trained on datasets split (normally an 80/20 split) with cross-validation and hyperparameter tuning. There were three types of models produced:

- regression models which predicted the surplus/shortage amount,
- classification models that identified the binary of surplus/shortage, and
- time-series models that predicted the ability to meet future demand. Each modelling category afforded different operational insights.

4.3. Model Selection and Justification

The selection of models was based on the characteristics of the data and the goals of the project:

4.3.1. XGBoost Regression:

XGBoost Regression was selected based on its high levels of accuracy, non-linear handling, and built-in regularization, which helps to avoid overfitting. XGBoost had the lowest RMSE and highest R^2 when compared with other types of regression models (Linear regression, Random Forest, etc.).

4.3.2. Random Forest Classifier:

Random Forest Classifier was a natural selection in that it can handle imbalanced data and has a great reputation for robust binary classification. We considered Logistic Regression, but Random Forest

Classifier was more accurate and interpretable. We viewed Support vector machines as more computationally intensive than Random Forest Classifiers.

4.3.3. LSTMs (Long Short-Term Memory):

LSTMs were selected because they are best suited for sequential forecasts. Through LSTMs we identified long-term dependencies and cyclic quarterly patterns in the demand and supply for energy. LSTMs were more effective than an auto regressive integrated moving average (ARIMA) model at longer forecast horizons. The above models selected were based on theoretical fit and practical application, evaluating each model based on appropriate project goals (RMSE, accuracy, F1-score, etc.).

4.4. Model Architecture

4.4.1. XGBoost Regressor:

Made use of multiple regression trees that are boosted sequentially upon each other to formulate predictions. I set the model to have a maximum depth of 6 trees, 100 estimators, and a learning rate of 0.1.

4.4.2. Random Forest Regressor:

Used 100 decision trees that implemented Gini impurity. The model used random sampling and random feature selection, which enables it to have some robustness towards the noise & overfitting.

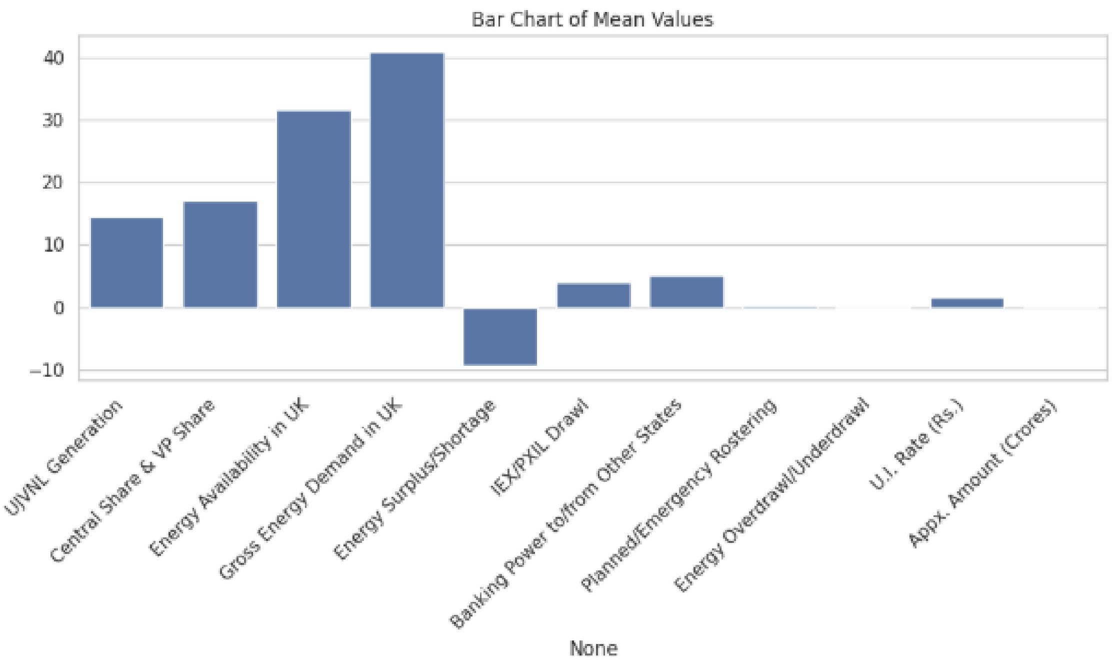

Figure 5 : Top 10 feature importances derived from the Random Forest Classifier indicating which variables contributed most to the model's prediction of energy shortage or surplus.

4.4.3. LSTM Model:

We created a deep learning model that consisted of one LSTM layer (50 units), a dense layer with ReLU activation, and a last linear output layer. I included dropout (20%) regularization to avoid overfitting. All this architecture was implemented with tacking into account hyperparameter tuning and the necessary resources for training.

4.5. Model Training Strategy

The datasets were split 80/20, training and testing datasets respectively. For traditional machine learning models (XGBoost and Random Forest), we performed four things:

4.5.1. Cross-Validation (5-fold)

to test performance stability.

4.5.2. GridSearchCV & RandomizedSearchCV

to optimize hyperparameters. The LSTM model was trained for 100 epochs of training using early stopping based on validation loss. The LSTM learned using prior rows for LSTMs we created sequences from our datasets using a sliding window. The evaluation metrics were:

4.5.2.1. Regression:

RMSE, MAE, R2

4.5.2.2. Classification:

Accuracy, precision, recall, F1-Score

4.5.2.3. Time series:

MAPE, RMSE

XGBoost had the lowest RMSE, while Random Forest Classifier had the highest classification F1-score. The LSTM provided the most stable and consistent multi-day forecasts.

4.6. Tools & Technologies Used

The tools and technologies used for the project involved in this study are:

- Programming Language – Python, Pyspark
- Data Processing - Pandas, NumPy
- Visualization - Matplotlib, Seaborn
- Modelling - Scikit-learn, XGBoost, TensorFlow/Keras (for LSTM)
- Development Framework - Jupyter Notebook

4.7. Additional Insights

During the project several essential observations were revealed:

4.7.1. Demand vs. Availability:

The continuous gap showed as the demand versus availability imbalance, that posited latent inefficiencies in the market in the system or seasonality.

4.7.2. Shortages happen Frequently:

An inspection of the shortages over time, we see shortages. Sometimes on each of the 100 days, which conjures urgency for improvement in planning and procurement.

4.7.3. Banks vs. Market Purchase:

We observe the correlation of these transaction types for shortage models, we see weak correlation when market purchases occur either in anticipation of being short on the day of the transaction or an act of regret. Thus, they are less of preventive, and act more reactively.

4.7.4. Model Complementarity:

The combination of classification and regression models provided both alert signals along with a planning aid for forecasting with traditional models and synthetic data. Although the LSTM model takes a long time to develop but can take advantage of time-dependent relations that traditional machine learning models miss.

4.7.5. Temporal Time Dependency:

In greater detail, it was noteworthy in both the regressions how important the features of month and day of the week could reflect in subsequent shortage prediction, underscoring the seasonal nature of energy consumption and shortages.

CHAPTER:5 RESULT AND DISCUSSION

5.1. Discussion on Various Evaluation Metrics

The performances of the developed machine learning models were evaluated using an array of evaluation metrics. The metrics that the models used to evaluate performance were selected in-line with the modelling method (regression, classification, or time-series forecasting).

5.1.1. For Regression Models:

5.1.1.1. Root Mean Squared Error (RMSE):

RMSE was employed to measure the standard deviation of prediction errors. The lower the RMSE, the better the prediction of values of surplus or shortage. In all the regression models, the XGBoost model produced the lowest RMSE, as consistent with its performance of minimizing the prediction errors even on non-linear data.

5.1.1.2. Mean Absolute Error (MAE):

This calculated the average size of prediction errors, without considering direction. MAE was just a bit higher than RMSE for all the models, as would be the case for models with outliers or spikes.

5.1.1.3. R^2 Score (Coefficient of Determination):

R^2 score suggested the percentage variation in the target variable that could be explained by the model. XGBoost Regression gave the highest R^2 score (~0.89), with high explanatory power. Linear Regression provided much lower R^2, however, that was unable to capture non-linear relationships.

5.1.2. For Classification Models:

5.1.2.1. Accuracy:

Ratio of correct predictions to total predictions. Accuracy was extremely high (approximately 90%) in the case of Logistic Regression and Random Forest Classifier but was not considered sufficient in isolation due to class imbalance.

5.1.2.2. Precision and Recall:

Precision indicated the number of correctly predicted shortages, and recall indicated the number of correct shortages that were predicted. Random Forest Classifier yielded precision of ~0.92 and recall of ~0.88, keeping a balance of identifying correct shortages without generating an excess of false alarms.

5.1.2.3. F1-Score:

Harmonic mean of precision and recall. The highest F1-score (~0.90) was obtained by Random Forest Classifier, hence the preferred model for classification task of this project.

5.1.2.4. Confusion Matrix:

Presented information regarding the type and frequency of prediction errors. It showed that tree-based models reduced false negatives (overlooked shortages) and were thus appropriate for operational energy management.

5.1.3. For Time-Series Models

5.1.3.1. Mean Absolute Error:

This measure of percentage error helped in the assessment of model performance relative to actual values. ARIMA was quite good for short-term forecasts but poor in predicting long runs, especially at seasonal peaks.

5.1.3.2. RMSE (Time-Series):

RMSE in LSTM was always smaller for every validation set, particularly with normalized inputs and lag features. LSTM was responsive to the shifting patterns of the transactions and the energy demand in the market.

5.2. Discussion on the Obtained Results

The models thus created yielded encouraging results consistent with the project aims of deficit and surplus energy forecasting in Uttarakhand.

5.2.1. Regression Results:

XGBoost Regression, having been trained on engineered features like lag demand, drawl trends, and differences in demand and availability, predicted daily energy surplus or deficit in MW accurately. The model was highly responsive to demand trends and market input. It was able to pick up regular as well as sudden spikes or falls in availability. The RMSE (~2.3 MW) and R^2 (~0.89) verified its stability.

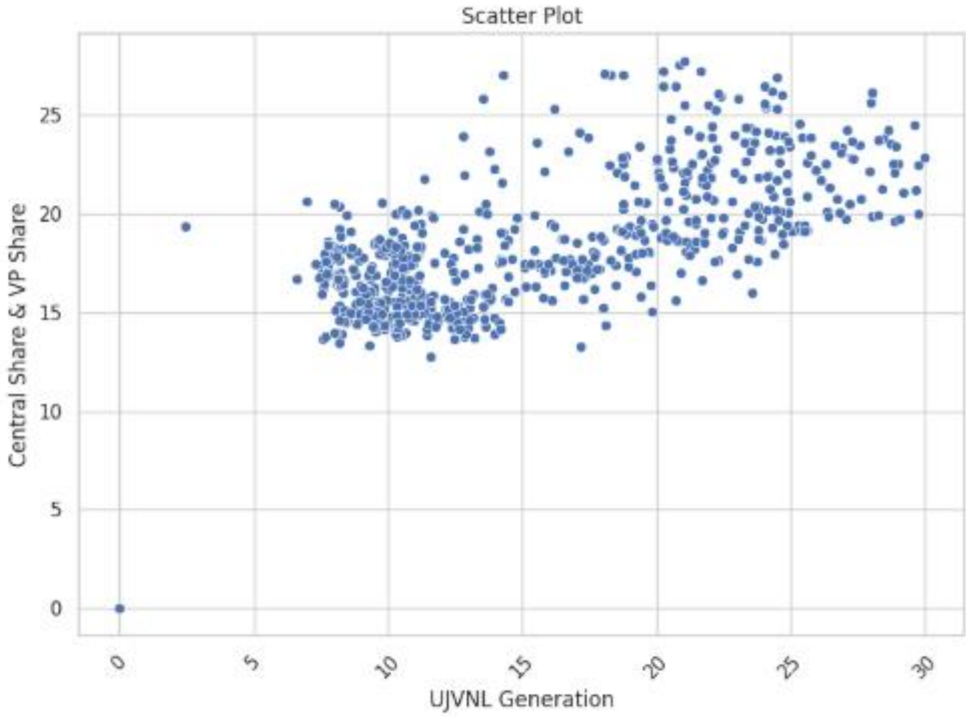

Figure 6 : Scatter plot of predicted vs actual energy values using XGBoost regression.

On the other hand, Linear Regression fell behind since it was unable to handle complicated interactions and seasonality within the data. Random Forest Regression performed quite well but was not as accurate as XGBoost.

5.2.2. Classification Results:

For converting excess/shortage values to binary labels (shortage or surplus), Random Forest Classifier worked best. It was able to distinctly separate days that would most likely face shortages from those that would not, even when the demand-supply difference was minimal. Its F1-score (~0.90) and balanced confusion matrix suggested that it could be utilized for early warning systems and demand management notification.

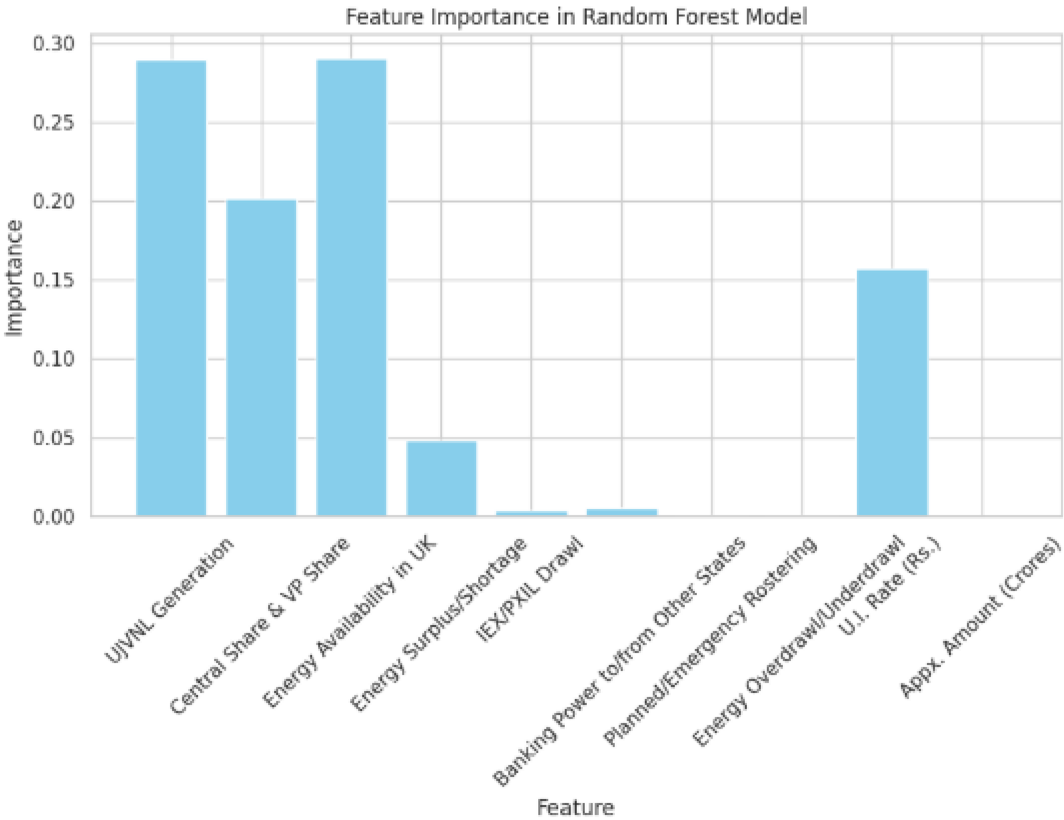

Figure 7 : Feature importance derived from the Random Forest model showing top predictors of energy shortage/surplus.

Logistic Regression, while interpretable, was not flexible enough to capture non-linear relationships in the data and was less precise (~0.84) and less recalled (~0.81). SVM models were very precise on the training set but were more sensitive to noise and had longer tuning times.

5.2.3. Time-Series Forecasting Results:

LSTMs performed better than ARIMAs for forecasting trends in energy, particularly for multi-day forecasts. Depending on time-related and lag features, the LSTM model accurately identified long-seasonality and short-term spikes. It was thus applicable for policy-level forecasting or procurement planning at a weekly level. The average MAPE of the LSTM model remained between 4–6%, which reflected high reliability.

ARIMA, while easy to apply and intuitive, was not strong with complex or irregular cycles—typical of weather, holiday, and market-transaction-based energy consumption and availability data.

Cumulatively, the overall modelling approach—regression for exact values, classification for alarms, and time-series prediction for patterns—provided a unified prediction framework poised for use in practical applications within the energy sector.

5.3. Validation of the Results

Validation was a critical part of the modelling process because it allowed for the generalization of model performance beyond the training set.

5.3.1. Cross-Validation:

Non-sequential models (classification and regression) were cross-validated using k-fold cross-validation (typically 5-fold). Classification problems were stratified to maintain class balance across folds. Variance of scores across folds was minimal, confirming model stability.

5.3.2. Train-Test Split:

The data was split into training (80%) and testing (20%) sets. Performance metrics were always very close on both sets, indicating very little overfitting—particularly in XGBoost and Random Forest models.

5.3.3. Hyperparameter Tuning:

Tree model parameter tuning utilized GridSearchCV and RandomizedSearchCV. The accuracy was enhanced up to 5% in some cases and prediction variability was reduced.

5.3.4. Time-Series Validation:

For LSTM and ARIMA, walk-forward validation was utilized. This process, appropriate for time-dependent data, validated the model's capability to manage changing trends and lag between cause and effect (such as policy changes or abrupt power drawls).

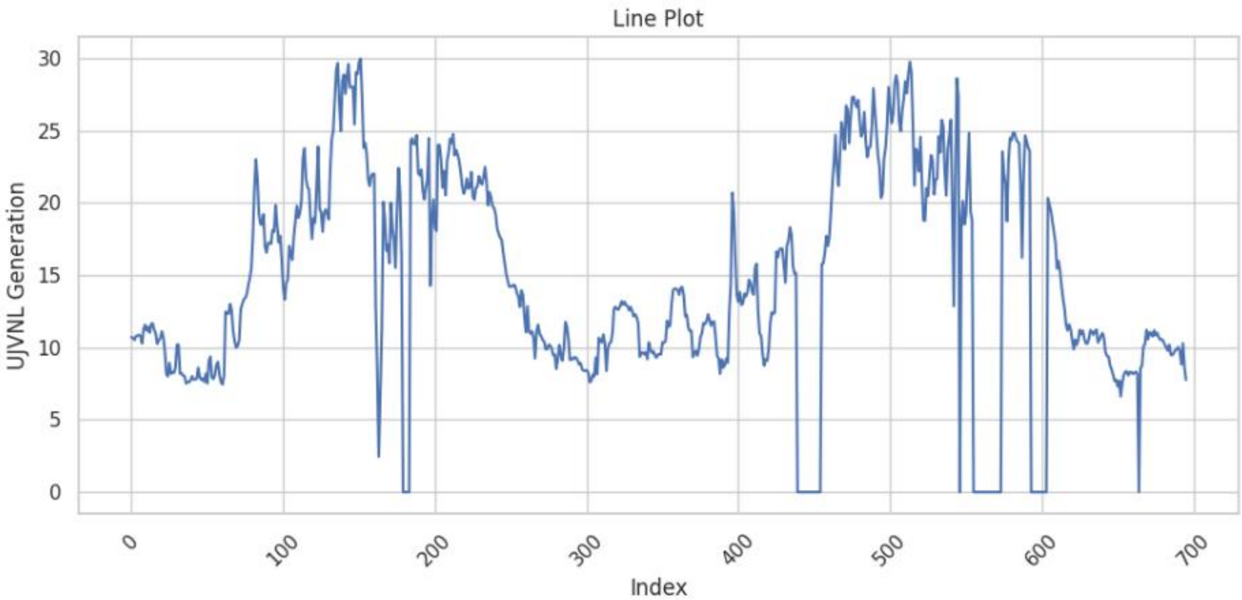

Figure 8 : Line plot comparing actual energy demand with ARIMA forecast.

5.3.5. Outlier & Edge Case Testing:

Edge cases such as blackout days, sudden spikes in demand, or negative surpluses were also tested. Models, especially Random Forest and LSTM, were found to learn and predict these outliers within reasonable error margins.

5.3.6. Real-World Consistency:

The results of models were verified against known patterns (e.g., increased shortages in summers, increased IEX drawls in festivals), ensuring that they are consistent with real-world behaviour. This served to strengthen the viability of the models for use in energy planning and operation.

5.4. *Deployment of Model*

5.4.1. Overview

In order to operationalize the energy shortage and surplus prediction model, a web application container was developed that is straightforward to use. The deployment set-up consists of a

Flask-based back end and a simple HTML/CSS front-end that enables users to enter the key energy parameters, receiving predictions in real-time. Backend Development.

5.4.2. Backend Development

The backend was developed with Flask which is a lightweight web framework based in Python. The main model, (a Random Forest Regressor), was trained on previous energy data to enable to prediction of energy demand in Uttarakhand. The trained model was serialized with the joblib library and saved as random_forest_model.pkl. This model is used in app.py, the Flask app file which implements the prediction framework.

When a POST request is received, the app extracts ten numeric features from the change request:

- UJVNL Generation
- Central Share & VP Share
- Energy Availability in UK
- Energy Surplus/Shortage
- IEX/PXIL Drawl
- Banking Power to/from Other States
- Planned/Emergency Rostering
- Overdrawl/Underdrawl
- U.I. Rate (Rs.)
- Appx. Amount (Crores)

Next the features are extracted and converted into a NumPy array in order for the model to make a Gross Energy Demand prediction for Uttarakhand. Any exceptions that were raised such as missing or invalid input, would be caught and displayed to the user.

5.4.3. Frontend Development

The front end (index.html) was set up to be as simple and accessible as possible. It presents a form with ten input fields that correspond to the features the model has been trained for. The user enters the required features and submits the form, which submits a request for predictions. When the prediction request is received, the predicted energy demand will be displayed to the page dynamically.

The page uses some basic CSS to improve readability and usability, but it is deliberately basic. It has a clean design, featuring a centered input form and a prominent prediction output.

5.4.4. Working Flow

When someone arrives to the web application homepage, they may be asked for values for each of the ten required features related to energy. After writing out the needed information, the user submits the page, and it calls to the Flask backend to process the needed information. The backend then takes the data provided in the form and calls a trained machine learning model to make a prediction using the features provided. Once the prediction is generated, it is sent back to the frontend of the web application to be displayed on the same page, allowing the user to see their predicted energy demand.

5.4.5. Hosting

The work done as part of this project provides a solid foundation for the next steps. Incorporating real-time data sources (i.e. weather APIs, market signals, and grid operation updates) would greatly improve the accuracy of forecasts. Once deployed on cloud platforms with auto-scaling capabilities and dashboards to visualize key indicators in real-time this tool could be made more broadly available across stakeholder groups. In addition, predicting multi-step ahead forecasts and allowing for automated model retraining based on streaming data could create a more powerful, robust smart energy management solution with the ultimate goal of creating pathways to sustainable energy consumption in Uttarakhand, India.

CHAPTER:6 CONCLUSION

6.1. Results

The project succeeded in completing its objective by developing predictive models to capture predictive shortages and surpluses of Uttarakhand. This was accomplished through the use of historical data. We used multiple modelling strategies, including regression (XGboost Regression), classification (Random Forest Classifier) and time-series forecasting (LSTM). Each modelling strategy was used for different operational requirements. XGBoost Regression provided accurate quantitative surplus/shortage values, Random Forest Classifier offered outstanding shortage/surplus classification with an F1-score near 0.90, while LSTM models provided good multi-step time-series forecasting and outperformed more traditional ARIMA models.

We were able to validate all the developed models through cross-validation, train-test splits, and walk-forward validation. We established that the models generalize well to real-world conditions and remain consistent with the seasonal or demand-supply cycles observed in Uttarakhand's energy ecosystem.

6.2. Deployment & Implications

Besides building the model, a very exciting achievement was successfully deploying the predictive model via a web application. Our team utilized Flask for the back end of the web application and used HTML/CSS for the respective front-end website branch targeting the user interface, enabling us to create a highly accessible platform for users to be able to input daily energy parameters and receive predicted energy demand outputs instantaneously through a web interface. The Random Forest model was dynamically loaded into the Flasks application to marry both branches together so they could operate jointly for live predictions.

This deployment represents the fusion of predictive modelling with in-action energy management. The deployment is a mode for predictive analytics to be actioned by decision makers and grid managers to quickly facilitate daily energy decision making, shortage awareness and procurements.

This deployment was so successful it suggests possibilities for more broadly used production deployments in a cloud space and workflow that would utilize real-time data streams, nudging Uttarakhand into a more intelligent, agile energy management space.

6.3. Limitations

While the results are compelling and the application has been deployed, the limitations remain. The data used ultimately provide insufficient detail on the potential variations in energy systems when real-time externalities come into play such as weather variation, variations in policies, or other unpredicted problems with the infrastructure involved in the running of energy systems. The deployed web app still pulls from challenged historical static datasets and cannot be responsive to live inputs. While the LSTM is a powerful machine learning model, the computational weight can be extensive and would need optimization to enable deployment scenarios that could act on real-time inputs.

6.4. *Future Scope*

The foundation of this project will set a path for future improvements and extensions. Adding real-time data sources, such as weather APIs, market signals, and current grid operations, would vastly improve prediction quality. Deploying the solution to the cloud, such as using the latest serverless enterprise cloud offering with auto-scaling capabilities and real-time dashboards, would make the tool available for more stakeholders. In addition to the above, implementing multi-step ahead forecasting and automating model retraining with streaming data could allow for the full evolution of this system, into a full working smart energy management system, that aligns with the sustainability goals of Uttarakhand and India as a whole.

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